

Introduction

Transporting heavy construction materials in space is prohibitively expensive, with an average cost of **\$10,000 per kilogram**. However, with NASA's Artemis missions, there is an increasing possibility of humans establishing a presence on the Moon; this means there is a **greater need for building shelter on the planet**. Hauling hundreds of pounds of concrete would require millions of dollars in funding, extra storage space aboard rocketry, and potentially multiple trips between the Earth and Moon. Considering NASA's reduced budget from 17 billion dollars to 15 billion this year, this is even less feasible. The Moon is already rich with a material that can be used to facilitate human settlement: **lunar regolith**, the soil that covers the entirety of the Moon's surface. **In situ resource utilization**, the use of materials already present on a planet to sustain human life, is an essential step towards developing a human habitat on the Moon.

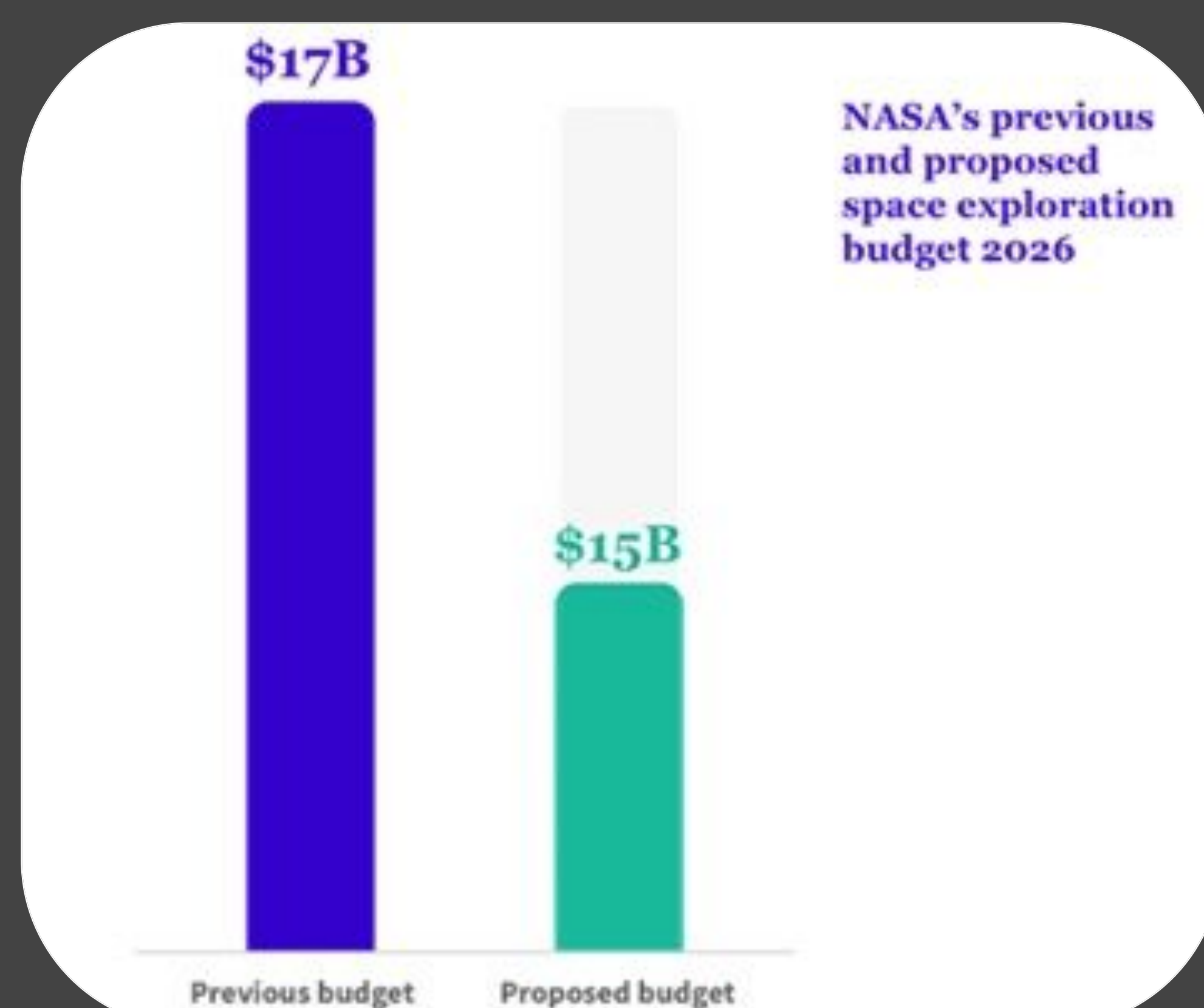


Figure 1. NASA Artemis Vehicle Expenditure

Purpose

Using lunar regolith, sustainable material must be synthesized for **habitat construction on the Moon**, which can be easily used through the development of lunar concrete 3D printers. By developing a **geopolymer concrete** capable of being 3D printed into structures on the Moon, the likelihood of a sustained human presence in space will dramatically increase.

In Situ Resource Utilization 3D Printing for Moon Habitation Methods

Printer Remodelling

- 3d printed housing, bracket, coupler, and 5 mm conical syringe tip
- Plunger mounted into housing (fig. 2)
- Rubber seal prevent leakage
- Lead screw and stepper motor connected to housing (fig. 3)
- 3 motors for all degrees of movement

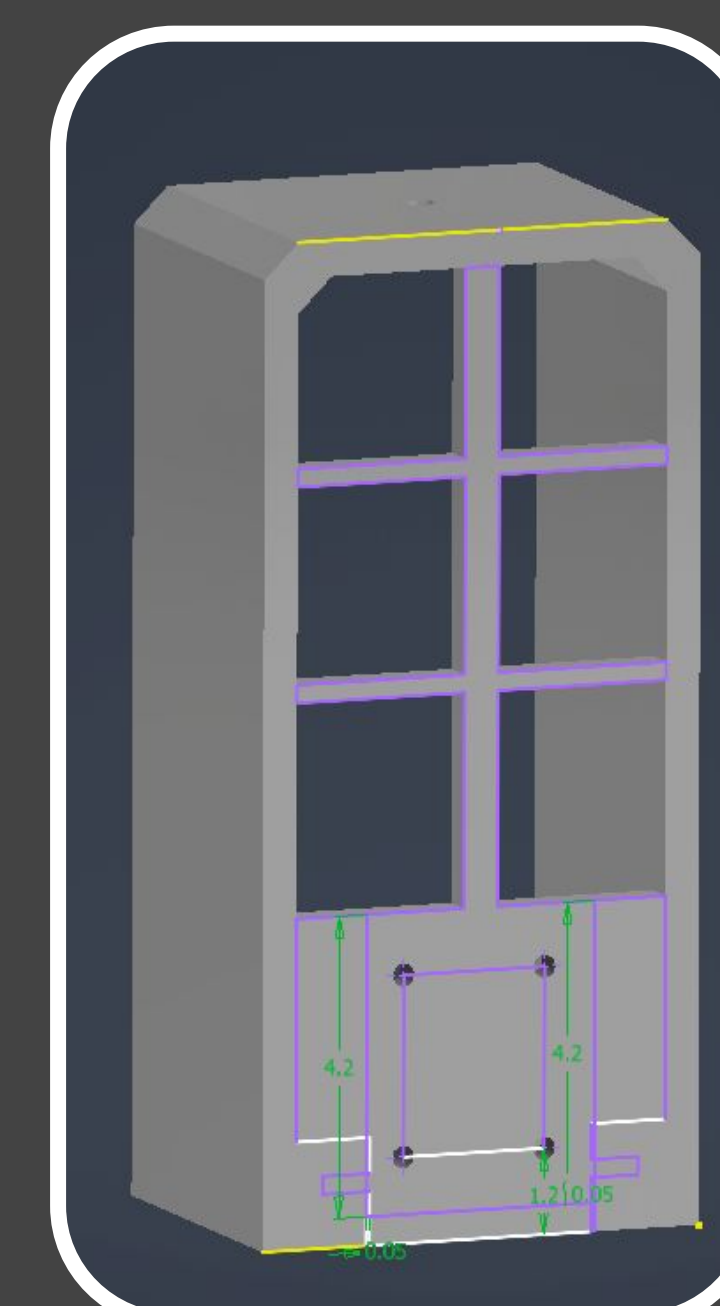


Figure 2. CAD of Housing

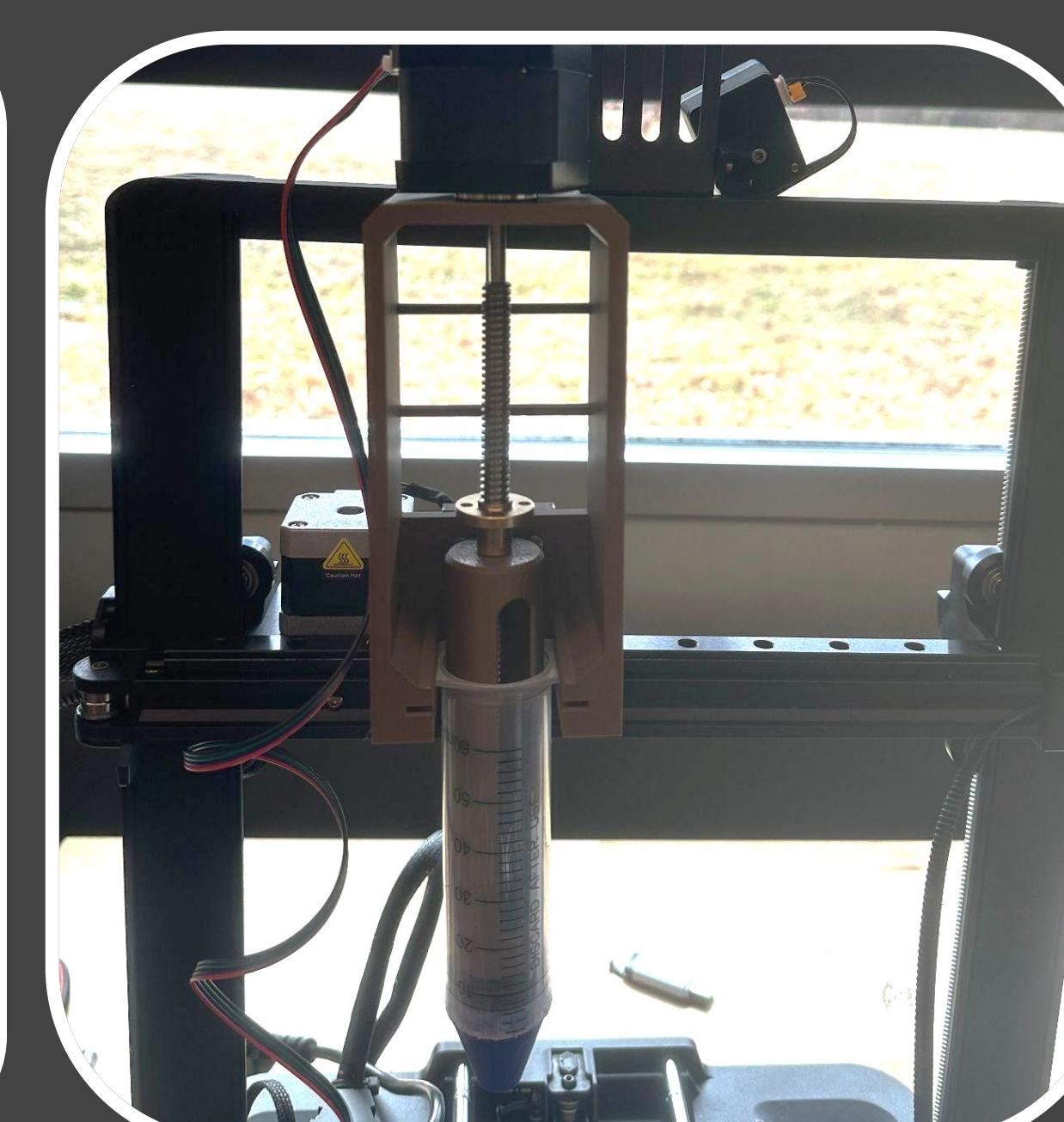


Figure 3. Extrusion System



Figure 4. Mixing Concrete

Geopolymer Concrete Formulation

- 8 M NaOH solution and sodium silicate activator solution
- 200g regolith simulant
- 1:3 and 2:3 sodium silicate to NaOH ratios tested in separate trials
- Final mixture was cohesive yet runny

Figure 2. Assembled Concrete Printer

3D Printing

- Syringe filled with mixture and tapped to remove air
- Syringe loaded into printer and lead screw rotated to extrude
- 1 x 1 inch puck samples (fig. 6)
- Samples were cured in an oven at 60 °C for 72 hours

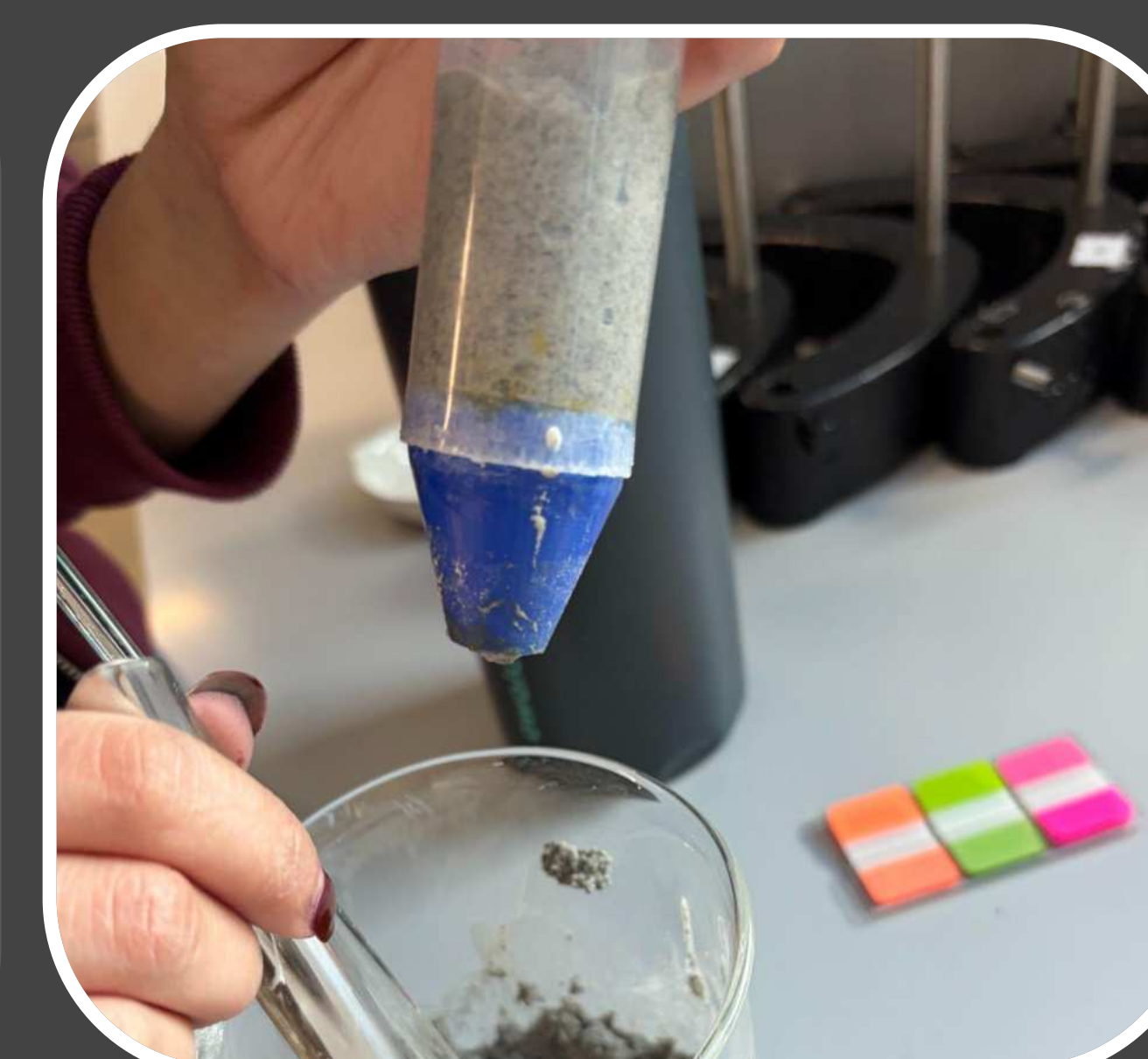


Figure 5. Syringe with Concrete



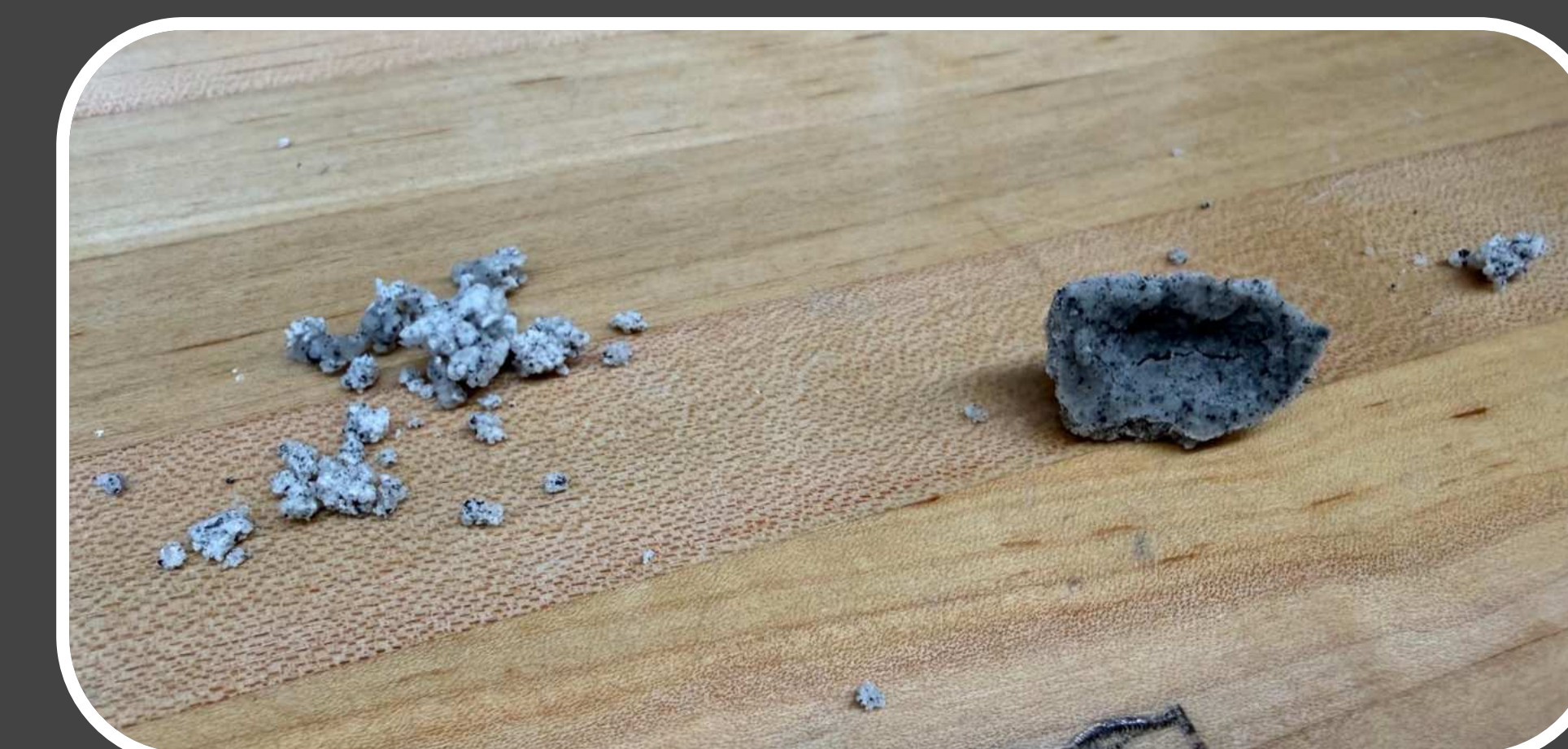
Figure 6. First Sample

Testing

- Sample height and circumference measured
- Compressive strength testing via 5kN universal testing machine
- Peak load used to calculate compressive strength

Results

Samples are still in the testing phase, meaning compression test data is not yet available. However, according to previous studies, the lunar concrete is projected to have a strength of **40 psi**. If the concrete is 20% or more weaker, the concrete formula will be readjusted in order to enhance strength.



Extensions

Beyond its primary use in constructing lunar habitats, in situ concrete printing systems can be used to develop large scale autonomous infrastructure on the Moon through robotic 3D printing. Prior to human arrival, structures such as landing pads, radiation resistant shelters, and storage units could be fabricated in situ in order to **reduce both mission risk and cost**. The geopolymer formulation can also be adapted or generalized for use on other planets and environments, such as Mars, where regolith based construction would similarly be necessary for sustained human presence. On Earth, the same approach has potential applications in **sustainable construction** by reducing reliance on traditional cement as well as enabling on site construction using locally sourced materials.

References

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- Pederson, F., Ellersick, L., & Kim, H.-J. (2025). A review of lunar regolith based alkali activated materials and sintered regolith for use as a construction material. *Acta Astronautica*, 232, 502–515. <https://doi.org/10.1016/j.actaastro.2025.03.032>